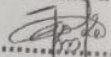


CSC 241: PYTHON

 COVENANT UNIVERSITY CANAAANLAND, KM 10, IDIROKO ROAD P.M.B 1023, OTA, OGUN STATE, NIGERIA.	COMPUTER & INFO. SCIENCES  HEAD OF DEPARTMENT Date: 6/2/23
B.Sc. DEGREE EXAMINATION	
COLLEGE: Science and Technology	DEPARTMENT: Computer & Info. Sciences
SESSION: 2022/2023	SEMESTER: ALPHA
COURSE CODE: CSC213	CREDIT UNIT: 3
COURSE TITLE: Structured Programming	TIME: 3 Hours
INSTRUCTION: Answer any FOUR (4) Questions	

Question One

a. Strings in Python are Immutable. Explain [2 Marks]

b. Complete the following programming exercises [2 Marks]

i. Given `full_name = 'Dr David Olaniyi Oyedepo'`, use String Slicing to output the Middle name [2 Marks]

ii. if `my_string = 'one two three four'` what will be the output of `word_list` from `word_list = my_string.split(o)` [2 Marks]

c. Identify the Errors in the following Code Segment [5 Marks]

```
2 string = "This Exam Is So Easy. My A Is Sure!"
3
4 # Converting to Lowercase
5 string_lower = lower(string)
6 print("Lowercase:", string_lower)
7
8 # Converting to uppercase
9 string_upper = upper(string)
10 print("Uppercase:", string_upper)
11
12 # Capitalizing the first Letter
13 string_capitalize = capitalize(string)
14 print("First letter capitalized:", string_capitalize)
15
16 # Title case
17 string_title = title(string)
18 print("Title case:", string_title)
19
20 # Reverse the string
21 string_reverse = string[::-0]
22 print("Reverse:", string_reverse)
```

d. Write a program that accepts a sentence as input and converts each word to modified "Goat Latin". In one version, to convert a word to Goat Latin, the following conditions apply:

i. If the word starts with a Vowel, then append 'MA' with the word

ii. If the word starts with a Consonant, then remove that from beginning, and append it at the end, then add 'MA' at the end.

iii. Add one letter 'A' to the end of each word

Here is an example: English: I SLEPT MOST OF THE NIGHT. Goat Latin: IMAA LEPTSMMAA OSTMMAA OFMAA HETMAA IGTNMAA. [6.5 Marks]

Question Two

a. Lists in Python are Ordered. Explain

b. Write the output of the following code snippets:

i. `myList = [[1, 2], [4, 5, 6]], [7, 8, 9]], [[10, 11], [13, 14, 15]]` [2 Marks]
`print (myList[0][1][2])`

ii. `numbers = [1, 2, 3, 4, 5]` [2 Marks]
`my_slice = numbers[-3:]*2`
`print(my_slice)`

c. Identify and correct the Error(s) in the following Code segment

```
1
2 shopping_list = ["apples", "bananas", "carrots"]
3 #print the 3rd item in the list
4 print(shopping_list[3])
5 shopping_list[4] = "dates"
6 shopping_list.remove("banana")
7 print(shopping_list)
8
```

[3 Marks]

d. The local driver's license office has asked you to create an application that grades the written portion of the driver's license exam. The exam has 20 multiple-choice questions. Here are the correct answers:

1. A	5. D	9. C	13. C	17. B
2. C	6. B	10. B	14. A	18. B
3. A	7. C	11. A	15. D	19. D
4. A	8. A	12. D	16. C	20. A

Your program should store these correct answers in a list. The program should read the student's answers for each of the 20 questions from with `input` function and store the answers in another list. After the student's answers have been read, the program should display the following: [8.5 Marks]

- A message indicating whether the student passed or failed the exam. (A student must correctly answer 15 of the 20 questions to pass the exam.)
- The total number of correctly answered questions
- The total number of incorrectly answered questions, and
- A list showing the question numbers of the incorrectly answered questions.

Question Three

a. Explain the concept of a KeyError in a Python Dictionary (Show relevant examples) [2 Marks]

b. What will be the Output of the following Code Snippets:

i. `my_dict = {"apple": "fruit", "carrot": "vegetable", "dog": "animal"}`
`my_dict["vegetable"] = "carrot"`
`print(my_dict)` [2 Marks]

ii. `months = {'Jan':1, 'Feb':2, 'Mar':3}`
`print(months[2])` [2 Marks]

c. The code is meant to add the Average score to the Dictionary. Identify the Errors in the following Code and Rewrite code with the Errors corrected. [5 Marks]

```

1 scores = {'Shade': [80, 75, 90], 'Bose': [75, 80, 85], 'Chima': [90, 80, 85]}
2
3
4 for student in scores:
5     avg_score = sum(scores[student])/len(scores[student])
6     scores[student] = avg_score
7
8 print(scores)

```

- d. Write a program that creates a dictionary containing 10 States in Nigeria as Keys, and their Capitals as Values. The program should then randomly quiz the user by displaying the name of a state and asking the user to enter that state's capital. The program should keep a count of the number of correct and incorrect responses. [6.5 Marks]

Question Four

- a. One property of Sets in Python is Uniqueness. Explain the concept of Uniqueness. [2 Marks]
b. What will be the output of the following code snippets [4 Marks]

i.	<pre> 1 myset = set([1, 2, 3]) 2 myset.update(['abc']) 3 print(myset) </pre>	ii.	<pre> 1 set1 = set([1, 2, 3, 4]) 2 set2 = set([3, 4, 5, 6]) 3 set1 = set1.union(set2) 4 print(set1) </pre>
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- c. Given the following code segment for capturing the population of a city in Nigeria: [5 Marks]

- i. Identify the **Error(s)** in the following codes and re-write the code in its correct form
ii. What are the outputs of the code?

```

2 population_data = {180,000,000, 170,000,000, 160,000,000}
3 print(population_data)
4 population_data.add(200,000,000)
5 population_data.remove(160,000,000)
6 print(population_data)
7 population_data.update({190,000,000, 200,000,000})
8 print(population_data)

```

- d. Write a program that reads the contents of two(2) Bible verses and use SET operations to perform the following comparisons: [6.5 Marks]
- Display a list of all the unique words contained in both verses.
 - Display a list of the words that appear in both verses.
 - Display a list of the words that appear in the first verses but not the second verse.
 - Display a list of the words that appear in the second verses but not the first verse.
 - Display a list of the words that appear in either the first or second verses, but not both.

Question Five

- a. Describe the concepts of Void Functions and Value-Returning Functions in Python (Show relevant examples) [3 Marks]
b. Given the definition for `my_function` [3 Marks]
- Write a statement that calls this function and uses keyword arguments to pass 'testing' into `x` and `2` into `y`.
 - What will be printed when the function call executes?

```

1 def my_function(x, y):
2     x=x*4
3     return x[y]

```

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c. Identify and correct the Error(s) in the following Code Segment

[4 Marks]

```

1 def order_food(item, quant):
2     food_price = {"pizza": 10, "burger": 5, "bread": 7}
3     price = 0
4     if item in food_item:
5         price = food_price[item] * quantity
6     else:
7         return "Sorry, that item is not available."
8     return price
9
10 food=input("Please Enter the food you want: ")
11 quantity=input("Enter the Quantity you want to buy: ")
12
13 food=food.lower()
14 bill = order_food (item,quant)
15
16 print(f'Your Bill is for {quantity} of {food.title()} is {bill}')

```

d. A computer hardware store sells various products such as laptops, desktops, and peripherals. The store must charge tax on the sale of these products and the tax rate varies depending on the product category. The tax rate for laptops is 12%, for desktops is 15%, and for peripherals is 10%. The store also offers a discount to its customers on certain products. For purchase more than N10000, the discount for laptops is 10%, for desktops and peripherals is 5%. Write a Python program that takes the product name, price, and category as input and calculates the final price after tax and discount. The program should use three functions: `final_price`, `compute_tax`, and `compute_discount`.

[6.5 Marks]

Question Six

a. Files in Python have Three(3) major access modes. Describe them.

[3 Marks]

b. Write the following short codes:

[3 Marks]

- A program that opens an output file with the filename `mythings.txt`, writes the your favorite Bible verse, Gospel Artist, and Christian Author to the file on separate lines, then closes the file
- A program that opens the `mythings.txt` file that was created in (i) reads all of the data from the file before closing it, and then displays the data from the file

c. Assume that a file named `scores.txt` exists on the computer's disk. It contains a series of records, each with two fields - a Coursecode, followed by a Score (an integer between 1 and 100). Write a program that displays:

[11.5 Marks]

- The number of records in the file
- The Course and Score of the record with the highest score
- The Average Score of all courses in the Record
- The Normalized Grade Point (NGP) of the Student based on the following formula:

$$NGP = \sum_{i=1}^n \frac{HighestScore - Score_i}{AverageScore}$$

Where **HighestScore** is the student's highest score from the file; The **AverageScore** is the Average score of all scores in the file and **Score_i** is the Score for each course.

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